Module/Pathway/Taxon Review: 2-Aminoethylphosphonate Degradation by PhnWX in *Pseudomonas putida* KT2440

Target taxon: *Pseudomonas putida* KT2440 (PSEPK; NCBI taxon 160488; proteome UP000000556) **Target pathway/bucket:** KEGG `ppu00440` — "Phosphonate and phosphinate metabolism" **Module area:** other_kegg_pathway (no module YAML resolved) **Candidate genes:** PP_2208 (*phnX*, Q88KT1), PP_2209 (*phnW*, Q88KT0)

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1. Executive Summary

The 2-aminoethylphosphonate (2-AEP, ciliatine) degradation module is **satisfiable and effectively complete** in *Pseudomonas putida* KT2440. The two canonical catabolic steps — transamination of 2-AEP and hydrolysis of the resulting phosphonoacetaldehyde — are encoded by an **adjacent, co-oriented *phnWX* gene pair**: PP_2209 (*phnW*, 2-aminoethylphosphonate:pyruvate transaminase, EC 2.6.1.37, KO K03430) and PP_2208 (*phnX*, phosphonoacetaldehyde hydrolase / "phosphonatase," EC 3.11.1.1, KO K05306). These two loci constitute the entirety of the KEGG `ppu00440` bucket for this strain. Both are Swiss-Prot reviewed, but their catalytic assignments rest on **HAMAP-rule inference (ECO:0000255)**, not direct KT2440 enzymology. Direct biochemical evidence for the identical two-enzyme route exists in other *P. putida* strains (NG2 and BIRD-1), which supports strong functional transfer to KT2440.

Critically, the **broad KEGG map `ppu00440` overstates the strain's true metabolic capability**. KT2440 degrades 2-AEP *exclusively* via the phosphonatase route. KEGG orthology queries confirm it **lacks the entire C–P lyase core** (*phnGHIJKLM*, KOs K06162–K06166, K05780–K05781), the **oxidative PhnY*/PhnZ pathway** (K21195/K21196), and phosphonoacetate hydrolase. These branches of the "Phosphonate and phosphinate metabolism" map are therefore **not encoded** in this organism and should be marked `not_expected_in_target_taxon` rather than treated as gaps.

For curation, the two core steps should be marked **covered**; the upstream **2-AEP uptake step is genome-encoded but ambiguous** (`candidate_uncertain`) — a phosphonate/phosphite ABC system (PP_0824–0827) and an ambiguously annotated putrescine-family ABC operon (PP_1722–1726) are both plausible importers, but no experimentally named dedicated AEP transporter (AepXVW/AepP/AepSTU) is annotated in KT2440. A divergent **LysR-family regulator, PP_2210**, adjacent to *phnWX*, is the candidate AepR ortholog. Finally, some broad HAD-superfamily GO terms attached to PP_2208 (phosphoglycolate phosphatase / DNA-repair activities) are **likely over-propagation** and should not be counted toward module satisfiability.

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2. Target-Organism Pathway Definition

2.1 What the process is

The relevant biochemical process is the **degradation of 2-aminoethylphosphonate (2-AEP)**, a naturally abundant phosphonate bearing a chemically stable C–P bond, into central metabolites plus inorganic phosphate. In KT2440 this proceeds by the **phosphonatase (substrate-specific) pathway**, a two-step route:

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              phnW (PP_2209)                        phnX (PP_2208)
              EC 2.6.1.37                           EC 3.11.1.1
2-AEP + pyruvate  ──────────►  phosphonoacetaldehyde  ──────────►  acetaldehyde + Pi
              + L-alanine         (+ H2O)
(RHEA:17021)                      (RHEA:18905)
PLP-dependent class-V            Mg2+-dependent HAD-superfamily
aminotransferase                 hydrolase; Schiff-base Lys
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The transaminase (PhnW) transfers the amino group of 2-AEP to pyruvate, yielding **phosphonoacetaldehyde** and L-alanine. The phosphonatase (PhnX) then hydrolytically cleaves the C–P bond of phosphonoacetaldehyde to release **acetaldehyde** and **inorganic phosphate**. Acetaldehyde feeds central carbon metabolism; Pi is assimilated. This route liberates phosphorus, carbon, and nitrogen from 2-AEP.

2.2 Pathway boundaries — what to keep separate

The KEGG map `ppu00440` "Phosphonate and phosphinate metabolism" is a **broad overview bucket** that aggregates several mechanistically distinct sub-pathways which should **not** be conflated for KT2440:

- **C–P lyase pathway** (*phnGHIJKLM* + accessory *phnCDE*, *phnNOP*): a broad-specificity, radical-SAM-based C–P cleavage machine used for many phosphonates including methylphosphonate. **Absent in KT2440.**
- **Oxidative 2-AEP pathway** (PhnY*/PhnZ): a non-heme Fe(II) oxygenase route converting 2-AEP to glycine + Pi, characteristic of certain marine bacteria. Absent in KT2440.**
- **Phosphonoacetate hydrolase** (*phnA*): degrades phosphonoacetate (a different substrate). **Not the KT2440 route.**
- **Phosphite/methylphosphonate oxidation** (ptxD etc.): reduced-P metabolism, mechanistically separate.

Only the **phosphonatase (PhnWX) branch** is instantiated in KT2440. Neighboring maps that should stay distinct include general **phosphate/phosphonate transport (ABC systems)**, **alanine/pyruvate metabolism** (downstream of the transamination), and the **PHO regulon** (phosphate-starvation response), which is a regulatory network rather than a catabolic pathway.

2.3 Alternate names and database definitions

- Substrate: **2-aminoethylphosphonate**, **2-AEP**, **2-AEPn**, **AEP**, **ciliatine**, **(2-aminoethyl)phosphonic acid**.
- *phnW* product: **2-aminoethylphosphonate—pyruvate transaminase / aminotransferase (AEPT / AEP transaminase)**, EC 2.6.1.37, KO K03430, HAMAP MF_01376.
- *phnX* product: **phosphonoacetaldehyde hydrolase / phosphonatase / phosphonoacetaldehyde phosphonohydrolase**, EC 3.11.1.1, KO K05306, HAMAP MF_01375.
- The operon as a system is referred to in the *Pseudomonas* literature as **"PhnWX"** (the transaminase–phosphonatase system).

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3. Expected Step Model

Step	Reaction	Enzyme (EC / KO)	KT2440 gene	Status
S0 (upstream)	2-AEP import across inner membrane	ABC transporter (various)	PP_0824–0827 and/or PP_1722–1726 (ambiguous)	`candidate_uncertain`
S1	2-AEP + pyruvate → phosphonoacetaldehyde + L-Ala	PhnW, EC 2.6.1.37 / K03430	**PP_2209**	**covered**
S2	phosphonoacetaldehyde + H_2O → acetaldehyde + Pi	PhnX, EC 3.11.1.1 / K05306	**PP_2208**	**covered**
R (regulation)	substrate-inducible transcriptional control	LysR-type (AepR)	PP_2210 (candidate)	`candidate_uncertain`
Alt-A	C–P lyase (broad phosphonate cleavage)	PhnGHIJKLM / K06162–K06166 etc.	—	`not_expected_in_target_taxon`
Alt-B	oxidative 2-AEP → glycine + Pi	PhnY*/PhnZ / K21195/ K21196	—	`not_expected_in_target_taxon`
Alt-C	phosphonoacetate hydrolysis	PhnA	—	`not_expected_in_target_taxon`

The **two-step phosphonatase core (S1 + S2) is fully covered**. Import (S0) and regulation (R) are present in the genome but require gene-name-level resolution. The three alternative branches are absent and expected to be absent.

4. Candidate Genes and Evidence

4.1 PP_2209 — *phnW* (2-AEP:pyruvate transaminase; Q88KT0)

Role. Catalyzes the committed first step: PLP-dependent transamination of 2-AEP with pyruvate to give phosphonoacetaldehyde + L-alanine (RHEA:17021). Classified by HAMAP MF_01376 as a class-V (fold-type I) aminotransferase.

Evidence type. Swiss-Prot reviewed; catalytic assignment via **HAMAP rule (ECO:0000255)** — i.e., high-confidence family inference, **not** a direct KT2440 assay. Strong functional support comes from *P. putida* strains where the identical activity was measured directly: in strain **NG2**, cell-free extracts contained "2-aminoethylphosphonic acid:pyruvate aminotransferase and phosphonoacetaldehyde hydrolase (phosphonatase) activities which were inducible by the presence of 2-aminoethylphosphonic acid" (PMID: 9841125). In *P. putida* **BIRD-1**, the operon is explicitly named "the 2AEP transaminase-phosphonatase system (PhnWX)" (PMID: 35229442).

Curation caveats. Because both direct-assay strains are *P. putida* (same species), transfer to KT2440 is **strong**. No known paralog competes for this KO in KT2440. Recommend **covered**.

4.2 PP_2208 — *phnX* (phosphonoacetaldehyde hydrolase / phosphonatase; Q88KT1)

Role. Catalyzes the second, C–P-bond-cleaving step: hydrolysis of phosphonoacetaldehyde to acetaldehyde + Pi (RHEA:18905). A **Mg^{2+}-dependent HAD-superfamily** hydrolase (HAMAP MF_01375). Located at genome coordinates ~2,514,559–2,515,386 (complement), immediately adjacent to *phnW* (PATRIC pegs 2330/2331).

Evidence type. Swiss-Prot reviewed with HAMAP-rule catalytic assignment. The **enzyme mechanism is directly established** in orthologs: recombinant phosphonatases from *Bacillus cereus* and *Salmonella typhimurium* were purified and shown to use a **Schiff-base lysine** mechanism. In *B. cereus*, "reaction … with phosphonoacetaldehyde or acetaldehyde in the presence of $NaBH_4$ resulted in complete loss of enzymatic activity" (PMID: 3132206); the *S. typhimurium* enzyme's mechanism "was verified … by phosphonoacetaldehyde-sodium borohydride-induced inactivation and by site-directed mutagenesis of the catalytic lysine 53" (PMID: 9649311). KT2440 PhnX belongs to the same HAMAP family, so the mechanism transfers with **high confidence** — but the direct assays were in non-*Pseudomonas* hosts, so this is **enzyme-family evidence, not a KT2440-specific assay**.

Curation caveats. As a HAD-superfamily member, PP_2208 may carry **over-propagated broad GO terms** (e.g., generic phosphoglycolate-phosphatase or DNA-repair "phosphatase" activities common to HAD annotations). These should **not** be counted toward module satisfiability; only the EC 3.11.1.1 phosphonatase activity is relevant. Recommend **covered**.

4.3 PP_2210 — LysR-family regulator (candidate AepR)

Role. Immediately downstream of *phnW*, forming a *phnX–phnW–regulator* neighborhood (PP_2208–PP_2209–PP_2210). This is the strong candidate **AepR ortholog**. In *P. putida* BIRD-1, a LysR-type regulator "termed AepR, upstream of the 2AEP transaminase-phosphonatase system (PhnWX)" governs induction (PMID: 35229442). Regulation is dual: global stress control acts "only … upon depletion of C, N, or P, controlled by CbrAB, NtrBC, or PhoBR respectively. However, the presence of 2AEP was necessary for full gene expression" (PMID: 35229442). Consistent with this, AEP catabolism in *P. putida* NG2 was substrate-inducible "regardless of the phosphate status of the cells" (PMID: 9841125) — i.e., **phosphate-starvation-independent**.

Curation caveats. The AepR assignment for PP_2210 is by synteny + homology, not direct KT2440 experiment. Mark `candidate_uncertain`.

4.4 Transport candidates (upstream step S0)

Two genome-encoded ABC systems are plausible for 2-AEP import; **neither is unambiguous**:

- **PP_0824–0827** — a phosphonate/phosphite ABC importer: PP_0824 *ptxB* (K02044, phosphonate substrate-binding protein), PP_0825 *phnC* (K02041, ATPase, EC 7.3.2.2), PP_0826 *phnE* (K02042, permease), PP_0827 *ptxC* (K02042, permease). Note: **no separate *phnD*/K02043** is present in this strain.
- **PP_1722–1726** — an ABC operon whose two permeases are GenBank-annotated "putative 2-aminoethylphosphonate transport system permease," yet whose KEGG KOs are **spermidine/putrescine-type (K02052–K02055)**, with an adjacent GntR regulator (PP_1727). This is a **conflicting / likely over-propagated annotation**.

No AepXVW/AepP/AepSTU ortholog is annotated by name in KT2440. The experimentally characterized dedicated 2AEP transporters were defined in marine bacteria: "we identify three bacterial 2-aminoethylphosphonate (2AEP) transporters, named AepXVW, AepP and AepSTU, whose synthesis is independent of phosphate concentrations" (PMID: 34315891). Because KT2440 lacks these by name, its AEP uptake route is **inferred/ambiguous** — mark `candidate_uncertain`.

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5. Gaps, Ambiguities, and Likely Over-Annotations

5.1 Confirmed absences (not gaps — genuinely not expected)

KEGG KO-linkage queries against organism `ppu` returned **ABSENT** for every C–P lyase core cistron: *phnG* (K06166), *phnH* (K05781), *phnI* (K05780), *phnJ* (K06162), *phnK* (K06163), *phnL* (K06164), *phnM* (K06165). The obligatory nature of these genes is established: "each of these strains contains phnGHIJKLM genes necessary in the C-P bond cleavage mechanism" (PMID: 31792787). Their absence means KT2440 **cannot** run the broad-specificity C–P lyase route. Similarly, the **oxidative pathway** genes *phnY (K21195) and* phnZ *(K21196) are ABSENT. The two degradative pathways are recognized as distinct and differentially distributed: "Homologous genes for both C-P lyase and phosphonatase degradative pathways are distributed in distantly related bacterial species" (PMID: 16245012). KT2440 possesses only the phosphonatase branch*.

Curation consequence: These branches of `ppu00440` should be scored `not_expected_in_target_taxon`, not `gap`. Failing to make this distinction would falsely flag the module as incomplete.

5.2 Ambiguities

- **2-AEP import (S0):** Two candidate ABC systems (PP_0824–0827; PP_1722–1726) but no named dedicated importer. The PP_1722–1726 "2-AEP permease" labels **conflict** with their spermidine/putrescine KO mapping. Genuine uncertainty.
- **Regulator (R):** PP_2210 is a strong-but-inferred AepR candidate.
- **Transporter completeness:** The PP_0824–0827 system lacks a separate *phnD* (K02043) substrate-binding component annotation, complicating confident assignment as a bona-fide phosphonate importer.

5.3 Likely over-annotations

- **PP_2208 broad HAD GO terms** (phosphoglycolate phosphatase / DNA-repair phosphatase activities): common HAD-superfamily over-propagation; not relevant to the AEP module.
- **PP_1722–1726 "2-aminoethylphosphonate transporter" name**: putative label at odds with KEGG polyamine-transporter orthology — likely over-propagated from a distant homolog.

6. Module and GO-Curation Recommendations

Module element	Recommended status	Rationale
S1 — 2-AEP transaminase (PP_2209 / PhnW)	**covered**	Swiss-Prot + HAMAP; direct activity in *P. putida* NG2/BIRD-1
S2 — phosphonatase (PP_2208 / PhnX)	**covered**	Swiss-Prot + HAMAP; direct mechanism in *B. cereus*/*S. typhimurium* orthologs
S0 — 2-AEP uptake	**candidate_uncertain**	Genome-encoded ABC systems, but importer identity ambiguous
R — AepR regulator (PP_2210)	**candidate_uncertain**	Strong synteny/homology, no direct KT2440 data
Alt-A — C–P lyase	**not_expected_in_target_taxon**	*phnGHIJKLM* KOs all absent
Alt-B — oxidative PhnY*/PhnZ	**not_expected_in_target_taxon**	K21195/K21196 absent
Alt-C — phosphonoacetate hydrolase	**not_expected_in_target_taxon**	Different substrate; not encoded

Module boundary assessment. The generic KEGG bucket `ppu00440` is **too broad** for accurate satisfiability scoring in KT2440. Recommendation: define/curate a **narrow "2-AEP phosphonatase (PhnWX) module"** capturing S1+S2 (core), with S0 and R as accessory steps. This yields a **fully satisfiable module** rather than an apparently partial `ppu00440` — i.e., the apparent incompleteness of `ppu00440` is a **module_needs_revision** artifact of overly broad boundaries, not a biological gap.

GO-curation actions: - Retain EC 3.11.1.1 (GO: phosphonoacetaldehyde hydrolase activity) for PP_2208; **flag/deprioritize** generic HAD phosphatase GO terms as likely over-propagation. - Retain EC 2.6.1.37 (GO: 2-aminoethylphosphonate–pyruvate transaminase activity) for PP_2209. - Flag the PP_1722–1726 "2-AEP transporter" annotation for review — its KEGG orthology (polyamine) conflicts with the name. - No new GO **term** requests appear strictly necessary (existing EC/GO cover the two steps), but a **new curated module document** for the PhnWX phosphonatase route is warranted.

7. Genes to Promote to Full `fetch-gene` Review

1. **PP_2208 (*phnX*)** — HIGH priority. Confirm the phosphonatase EC assignment and audit/strip over-propagated HAD GO terms.
2. **PP_2209 (*phnW*)** — HIGH priority. Confirm transaminase assignment; establish covered status with citation to *P. putida* NG2/BIRD-1 activity data.
3. **PP_2210 (LysR regulator)** — MEDIUM. Resolve whether it is the AepR ortholog controlling the operon.
4. **PP_1722–PP_1726 (ABC operon)** — MEDIUM. Resolve the name-vs-KO conflict (2-AEP permease label vs spermidine/putrescine KOs); determine whether it contributes to 2-AEP import.
5. **PP_0824–PP_0827 (ptxB-phnC-phnE-ptxC)** — LOW/MEDIUM. Clarify whether this phosphonate/phosphite ABC system imports 2-AEP and why *phnD* is missing.

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8. Mechanistic Model and Interpretation

Synthesizing the five confirmed findings yields a coherent, species-aware picture:

```
Extracellular 2-AEP
       │
       │  S0: ABC import (candidate: PP_0824–0827 phosphonate/phosphite
       │      ABC, or PP_1722–1726 ambiguous ABC operon)   [candidate_uncertain]
       ▼
Cytoplasmic 2-AEP
       │
       │  S1: PhnW / PP_2209  (EC 2.6.1.37, K03430)         [COVERED]
       │      + pyruvate → + L-alanine
       ▼
Phosphonoacetaldehyde
       │
       │  S2: PhnX / PP_2208  (EC 3.11.1.1, K05306)          [COVERED]
       │      Schiff-base Lys / HAD hydrolase; + H2O
       ▼
Acetaldehyde  +  Pi  →  central metabolism / P assimilation

Regulation: LysR AepR candidate PP_2210 — substrate (2AEP)-inducible,
            phosphate-starvation-independent, layered on CbrAB/NtrBC/PhoBR

NOT ENCODED in KT2440:  C–P lyase (phnGHIJKLM) · oxidative PhnY*/PhnZ · phnA
```

The **operon architecture is decisive**: PP_2208 (*phnX*) and PP_2209 (*phnW*) are immediately adjacent, and together they are the *only* two genes KEGG maps to `ppu00440` in this strain. The presence of a divergent LysR regulator (PP_2210) directly abutting the operon matches the experimentally defined AepR/PhnWX arrangement in *P. putida* BIRD-1, strengthening the interpretation that this is a genuine, regulated catabolic operon rather than a set of orphan homologs.

The **enzymology is well understood at the family level**. The PhnW transamination is a standard PLP-dependent reaction; the PhnX step is the mechanistically interesting one — a HAD-superfamily hydrolase that cleaves the exceptionally stable C–P bond via a covalent Schiff-base intermediate on a catalytic lysine, assisted by a conserved active-site aspartate. This mechanism was proven directly in *B. cereus* and *S. typhimurium* orthologs, and because KT2440 PhnX is in the same HAMAP family (MF_01375), the assignment carries high confidence. A QM/MM study (PMID: 18802516) offers a proton-transfer variant of the mechanism, but both models agree the reaction is a Lys/Asp-dependent HAD hydrolysis.

The **species-transfer logic** is central to curation. Direct enzyme activity for the two-step phosphonatase route was demonstrated in *P. putida* NG2 (same species), making functional transfer to KT2440 **strong**; the KT2440-specific catalytic evidence itself is HAMAP inference.

The absence calls for the C–P lyase and oxidative branches are based on KEGG orthology, which is consistent with the biology (KT2440 is a soil pseudomonad using the substrate-specific phosphonatase route, not the broad C–P lyase machine).

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9. Evidence Base

PMID	Organism / Scope	Supports
9841125	*P. putida* NG2 (same species)	**Direct** AEP:pyruvate aminotransferase + phosphonatase activities; substrate-inducible, phosphate-independent. Strongest functional transfer to KT2440 (F001, F003).
35229442	*P. putida* BIRD-1 (same species)	PhnWX operon + LysR regulator AepR; dual regulation (CbrAB/NtrBC/PhoBR + 2AEP inducer). Model for PP_2210 (F001, F003).
9649311	*S. typhimurium* (family)	Schiff-base Lys53 mechanism of phosphonatase verified by $NaBH_4$ inactivation + mutagenesis. Underpins EC 3.11.1.1 for PhnX (F005).
3132206	*B. cereus* (family)	Original active-site Schiff-base lysine evidence for phosphonatase (F005).
31792787	Review / general	Establishes *phnGHIJKLM* as obligatory C–P lyase core; basis for absence call (F002).
16245012	Review / general	Distinguishes C–P lyase vs phosphonatase degradative pathways and their distribution (F002).
34315891	Marine bacteria	Defines dedicated 2AEP transporters (AepXVW/AepP/AepSTU); KT2440 lacks these by name → uptake ambiguous (F004).
18802516	Computational	QM/MM alternative proton-transfer mechanism for phosphonoacetaldehyde hydrolase (mechanistic nuance).
30810303	*Gimesia maris*	Oxidative PhnY*/PhnZ pathway characterization (branch absent in KT2440).

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10. Limitations and Knowledge Gaps

- **No KT2440-specific enzymology.** Both PhnW and PhnX catalytic roles are HAMAP-rule inferences (ECO:0000255). Direct activity is demonstrated in *other P. putida* strains (NG2, BIRD-1) and the phosphonatase mechanism in *B. cereus*/*S. typhimurium* — strong but indirect for KT2440 specifically.
- **Uptake unresolved.** The physiological 2-AEP importer in KT2440 is not experimentally identified; two conflicting candidate systems exist and no named dedicated importer is annotated.
- **Regulation inferred.** PP_2210 as AepR is by synteny/homology; no KT2440 transcriptomic confirmation.
- **KEGG-orthology-based absence calls.** The C–P lyase and PhnY*/PhnZ absences rest on KEGG KO linkage; while consistent and biologically sensible, a targeted genome-wide HMM search could formally exclude highly diverged homologs.

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11. Proposed Follow-up Experiments / Actions

1. **Growth phenotyping:** Test KT2440 for growth on 2-AEP as sole P (and N/C) source under phosphate-replete and -deplete conditions; expect substrate-inducible, phosphate-independent catabolism (as in NG2).
2. **Targeted knockouts:** ΔPP_2208 and ΔPP_2209 should abolish 2-AEP utilization; ΔPP_2210 should impair induction — confirming module roles.
3. **Enzyme assays:** Direct measurement of AEP:pyruvate transaminase and phosphonatase activities in KT2440 cell-free extracts to convert HAMAP inference to direct evidence.
4. **Transporter resolution:** Knock out PP_1722–1726 and PP_0824–0827 individually and in combination; measure labeled 2-AEP uptake to identify the physiological importer.
5. **Transcriptomics/reporter fusions:** Confirm PP_2210-dependent, 2-AEP-inducible expression of the *phnWX* operon.
6. **Curation actions:** Create a dedicated PhnWX phosphonatase module; mark S1/S2 covered, S0/R candidate_uncertain, and the C–P lyase / oxidative / phosphonoacetate branches not_expected_in_target_taxon; flag PP_2208 broad HAD GO terms and the PP_1722–1726 name as over-propagations; promote PP_2208 and PP_2209 to full `fetch-gene` review.

Bottom line for curators

The **PhnWX 2-AEP phosphonatase module is covered and satisfiable in *P. putida* KT2440** via the adjacent PP_2209 (*phnW*) / PP_2208 (*phnX*) pair. The broad `ppu00440` bucket should be **narrowed**: mark the two core steps covered, uptake and regulation candidate_uncertain, and the C–P lyase, oxidative PhnY*/PhnZ, and phosphonoacetate branches not_expected_in_target_taxon. Watch for HAD-superfamily GO over-propagation on PP_2208 and a name/KO conflict on the PP_1722–1726 "2-AEP transporter."
